

Agenda.

→ Self Reflection

→ Feedback loops.

→ Writing an email to your manager

Self-Reflection means that AI model checks & evaluates the response itself.

Feedback loop means that AI model uses the feedback or improvement pointers from user, agent, tool or environment to improve future actions.

Agents.

Perception → Reasoning → Action.

⇒ Self-Reflection is the process where an AI agent evaluates its own reasoning, planning, action & then tries to improve it.

⇒ Feedback loop happens when the result of an action is given back to the system so that the system can its future actions.

GPT-0SS

Chat Bot Assistant.

⇒ A feedback loop is a cycle where an agent performs an actions, observes the results, gets the feedback & will try to improve in the future.

⇒ Iterative feedback loop

Why Single shot AI response is NOT good?

→ Ask a question to LLM

→ get the response.

May not be
100% accurate

↓
Incomplete

→ Some failure.

⇒ Without self-reflection & feedback loops, Agents can go wrong

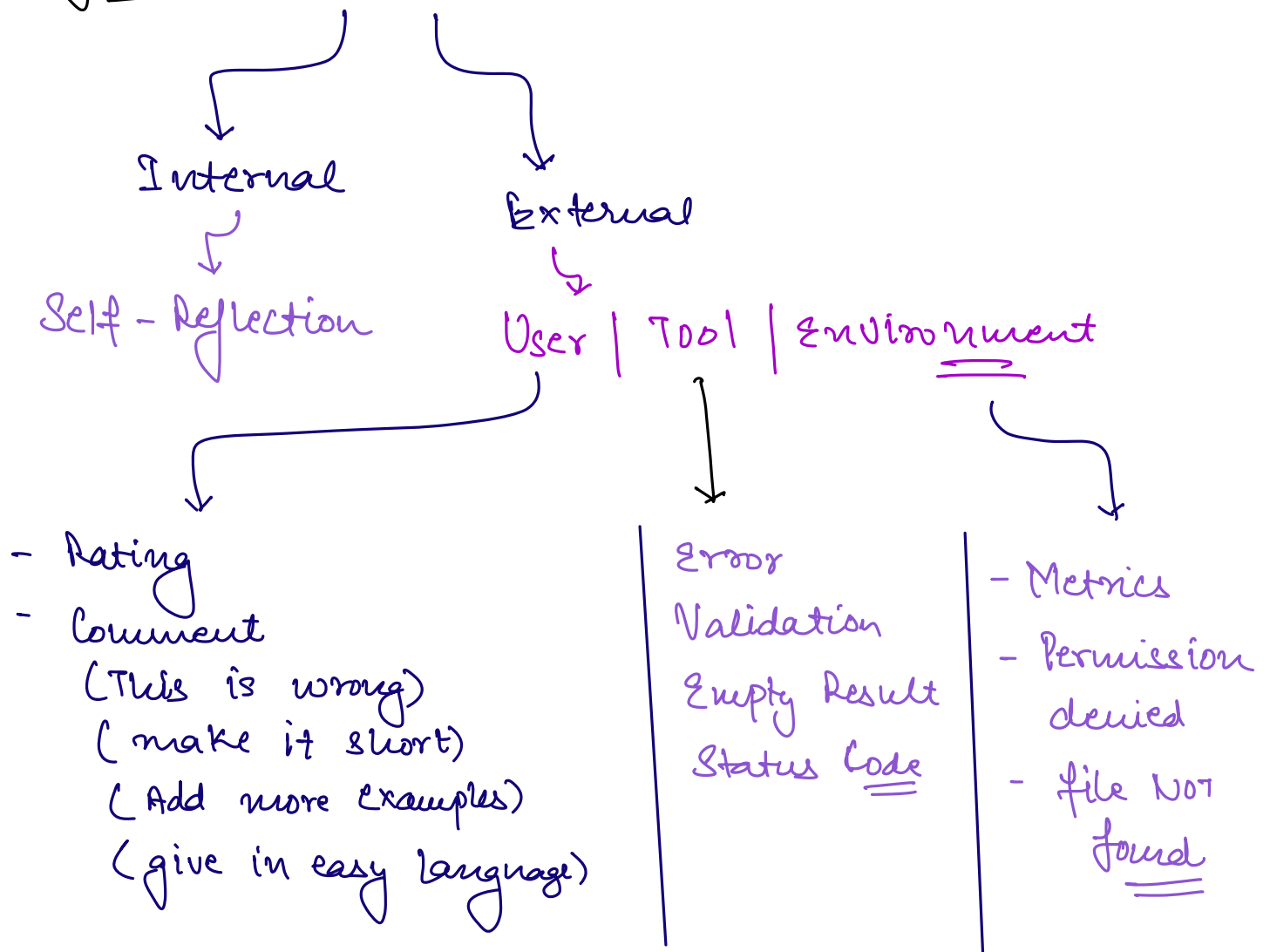
⇒ That's why self-reflection & feedback loops are very important for building real-world production reliable agents.

→ Benefits of Self-Reflection

- Accuracy
- Completeness
- Safety & security
- Better user experience

Goal → Plan → Reflect → Observe → Improve
↓
Act.

Type of feedbacks.



Self - Reflection Patterns.

1. (Plan)

Draft → Review → Revise

Ex: Chat Bot Assistant.

Content writing

Travel Booking agent.

2.

Plan → Execute → Check → Repan

Ex: Resume Write

Automation.

3. Tool Based Reflection

↪ Trigger the reflection based on tool's response / failure.

Self-Reflection means infinite thinking?

→ we need to find the right balance b/w
response quality (vs) self-reflection / feedback.

→ Too much self-reflection can :

i) increase latency

ii) Cost

iii) resources (time | power | tokens...)

iv) hamper user experience.

Case Study

→ Customer Support Agent.

Query: I ordered shoes, but I received the
wrong size.

Agent design

Step I: Understand the issue:

- Order issue
- Order Id
- Wrong size delivered.

Step II: Reasoning / Planning

- Validate orderId
- Validate size delivered
- Check return / replacement / refund policy

Step III: Self-Reflection

- Did we check the correct?
- Politeness
- did we check the right policy?

Step IV : Response

Step V : Feedback.

Case Study

- Travel Planning agent
- Coding Agent

Reflection via Prompts.

[Prompt] + Please review & validate
the response before returning.

- Identify weakness in the previous response.
- Suggest improvements in the previous response